

JEE MAIN REVISION MASTER: 3D GEOMETRY SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 385 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Planes & Line-Plane Intersections in 3D	236 PYQs	3D Planes
Very High	Direction Cosines & Equations of Lines in 3D	156 PYQs	3D Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Planes & Line-Plane Intersections in 3D	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, $d = \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$, $\sin\theta = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$
Direction Cosines & Equations of Lines in 3D	$l^2 + m^2 + n^2 = 1$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$, $d_{\text{skew}} = \frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Planes & Line-Plane Intersections in 3D (Very High Yield — 236 PYQs)

Core Principle: Normal vector to a plane, perpendicular distance from point to plane, and angle between a line and plane (uses $\sin\theta$).

■■ **Caution / JEE Pitfall:** Angle between a LINE (vector $\vec{b}$) and a PLANE (normal $\vec{n}$) is computed with SINE ($\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$), NOT cosine!

■ **Speed Trick:** A line is parallel to a plane if and only if it is perpendicular to the plane's normal: $\vec{b} \cdot \vec{n} = 0$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $(x-4)/1 = (y-9)/2 = (z-5)/1$ be the point $(\lambda, \mu, 2\dots)$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let L be the line $(x+1)/2 = (y+1)/3 = (z+3)/6$ and let S be the set of all points (a, b, c) on L , whose distance from the li..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "If $(2\alpha+1, \alpha^2-3\alpha, (\alpha-1)/2)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{\dots}$ " [Ans: N/A]

3.2 Direction Cosines & Equations of Lines in 3D (Very High Yield — 156 PYQs)

Core Principle: Direction cosines (l, m, n), line passing through two points, and shortest distance formula between skew lines.

■ **Caution / JEE Pitfall:** Direction ratios (a, b, c) can be any proportional numbers, but direction cosines (l, m, n) MUST satisfy $l^2 + m^2 + n^2 = 1$.

■ **Speed Trick:** Two lines in 3D intersect (are coplanar) if and only if the shortest distance between them is ZERO: $(\text{vec}(a)_2 - \text{vec}(a)_1) \cdot (\text{vec}(b)_1 \times \text{vec}(b)_2) = 0$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Let a triangle PQR be such that P and Q lie on the line $(x+3/8)=(y-4/2)=(z+1/2)$ and are at a distance of 6 units from R..." [Ans: C]
- **[2026 Evening Shift] Q#1 (MCQ):** "The shortest distance between the lines $\text{vec}(r)=((1/3)i+2j+(8/3)k)+\lambda(2i-5j+6k)$ and $\dots$ " [Ans: B]
- **[2026 Evening Shift] Q#1 (MCQ):** "Let the point A be the foot of perpendicular drawn from the point P ($a,b,0$) on the line $(x-1/2)=(y-2/1)=(z-\alpha/3)$" [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let a triangle PQR be such that P and Q lie on the line $(x+3/8)=(y-4/2)=(z+1/2)$ and are at a distance of 6 units from R(1,2,3). If (α, β, γ) is the centroid of Δ PQR, then $\alpha + \beta + \gamma$ is equal to :

- (A) 4
- (B) 5
- (C) 6 [Correct]
- (D) 8

Correct Answer: C

Solution Analysis: Write the given line in parametric form. $(x+3/8)=(y-4/2)=(z+1/2)=t$ So any point on the line is $P(t)=(-3+8t, 4+2t, -1+2t)$ Since both P and Q lie on this line and are at a distance 6 from R(1,2,3), we find the points on the line whose distance from R is 6. Distance condition: $\sqrt{(x-1)^2+(y-2)^2+(z-3)^2}=6$ Substitute $x=-3+8t, y=4+2t, z=-1+2t$: $(-3+8t-1)^2+(4+2t-2)^2+(-1+2t-3)^2=36$ $(8t-4)^2+(2t+2)^2+(2t-4)^2=36$ Now expand: $64t^2-64t+16+4t^2+8t+4+4t^2-16t+16=36$ $72t^2-72t+36=36$ $72t^2-72t=0$ $72t(t-1)=0$ So, $t=0$ or $t=1$

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Evening Shift (Q#2)

If the image of the point P($a, 2, a$) in the line $(x/2)=(y+a/1)=(z/1)$ is Q and the image of Q in the line $(x-2b/2)=(y-a/1)=(z+2b/-5)$ is P, then $a+b$ is equal to ____.

Correct Answer: 3

Solution Analysis: For line 1: $(x/2)=(y+a/1)=(z/1)=\lambda$ general point M on line is $(2\lambda, \lambda-a, \lambda)$ P is ($a, 2, a$) $\text{vec}(\text{PM})=(2\lambda-a, \lambda-a-2, \lambda-a)$ since PM is perpendicular to line 1, their dot product is 0 $\text{vec}(\text{PM}) \cdot (2, 1, 1)=0$ $2(2\lambda-a)+1(\lambda-a-2)+1(\lambda-a)=0$ $6\lambda-4a-2=0 \Rightarrow \lambda=(2a+1/3)$ midpoint M = $((4a+2/3), (1-a/3), (2a+1/3))$ for line 2: $(x-2b/2)=(y-a/1)=(z+2b/-5)$ since P is the image of